Recognizing IIC-comparability graphs

Marina Groshaus André Guedes

UTFPR UFPR

Combinatorics seminars in Rio de Janeiro

2026

Bicliques

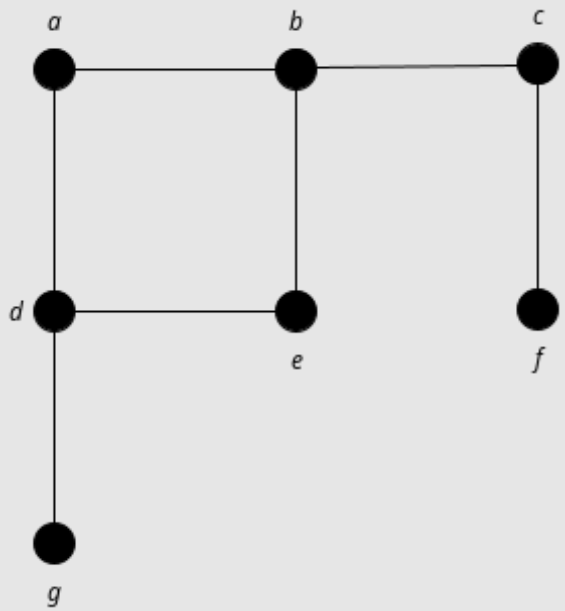
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A *biclique* of a graph G is a set B of vertices of G such that B induces a maximal complete bipartite graph.

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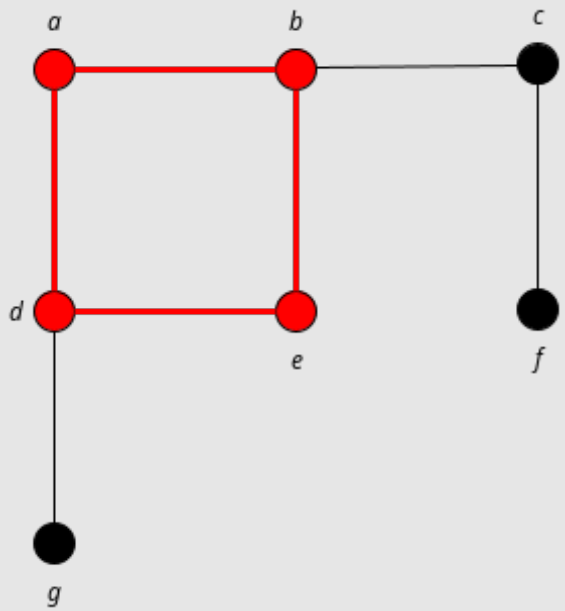
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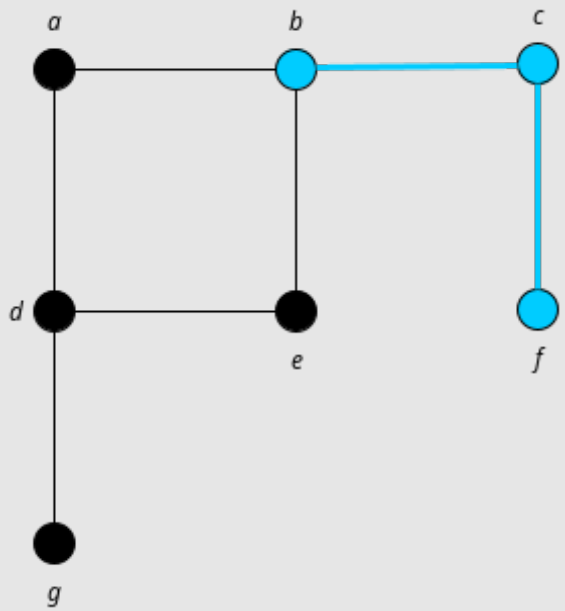
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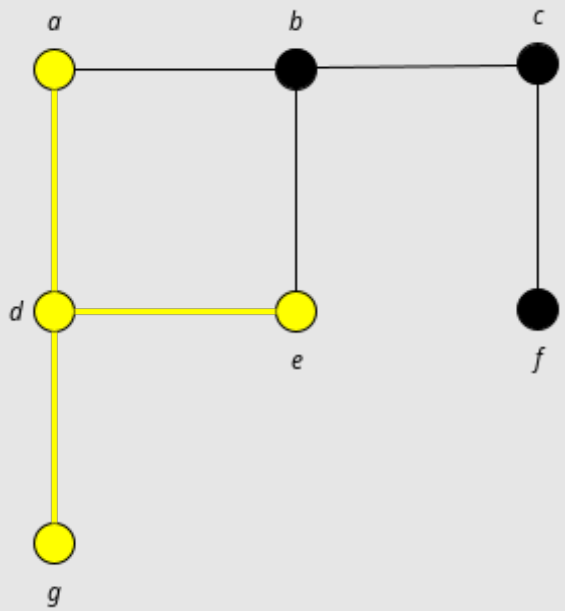
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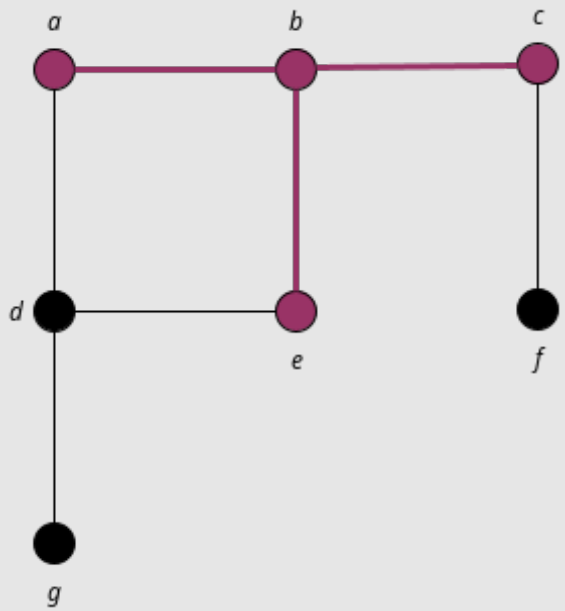
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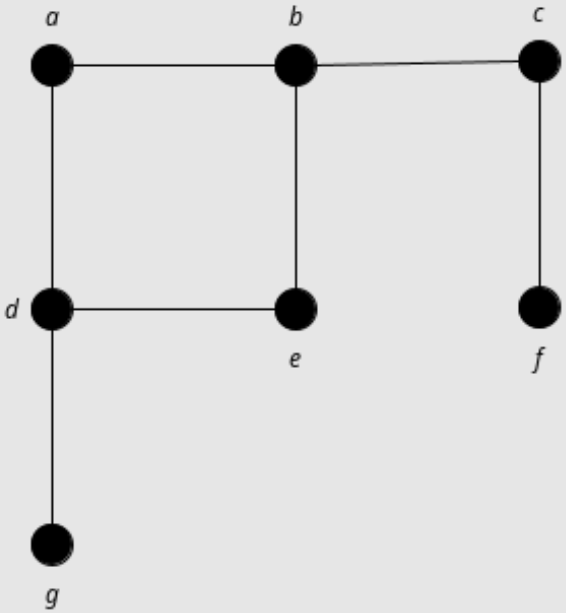
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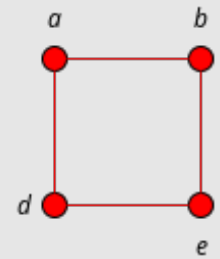
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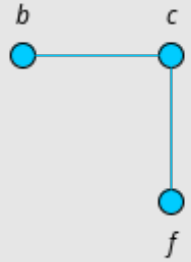
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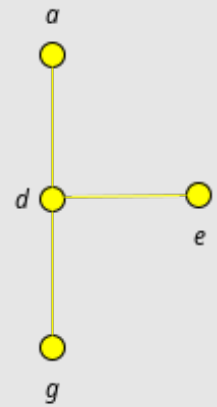
G

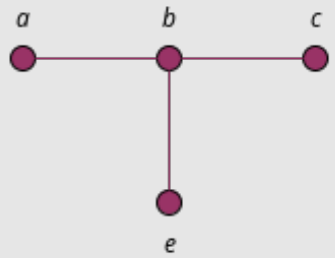


A

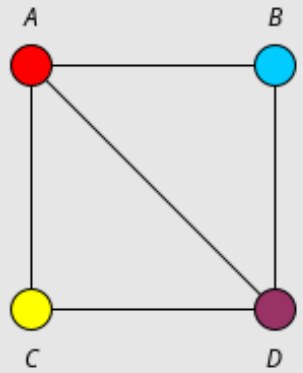


B





D



KB(G)

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- The characterization does not lead to a polynomial time recognition algorithm.
- Since then, it remains open the time complexity of the problem of recognizing biclique graphs.

IIC-Comparability graphs

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$KB(\text{bipartite}) = (IIC\text{-comparability})^2$ [Groshaus, Guedes (2021)].

But, what do we know about the *IIC*-comparability class?

Comparability graphs

$G(\mathcal{P}) = (C, E)$ is the *comparability graph* of a poset $\mathcal{P} = (C, \leq)$ when:

- $uv \in E \iff u$ and v are *comparable* in $\mathcal{P}$

Comparability graphs

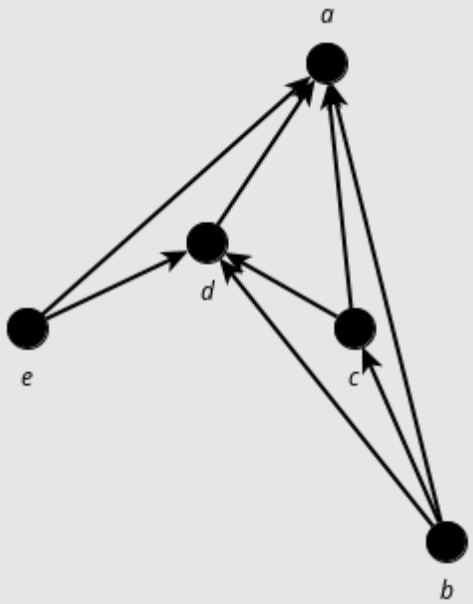
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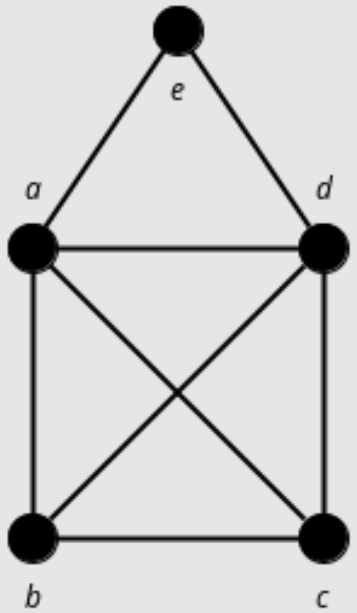
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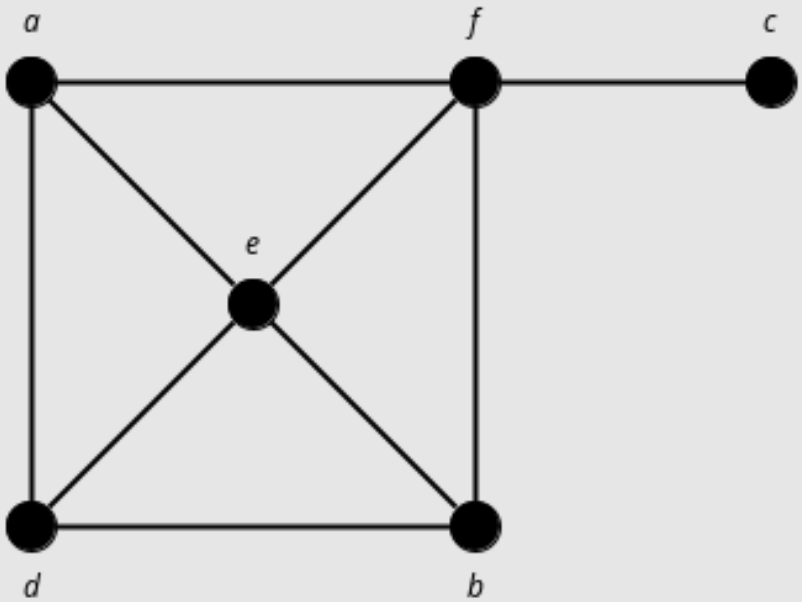


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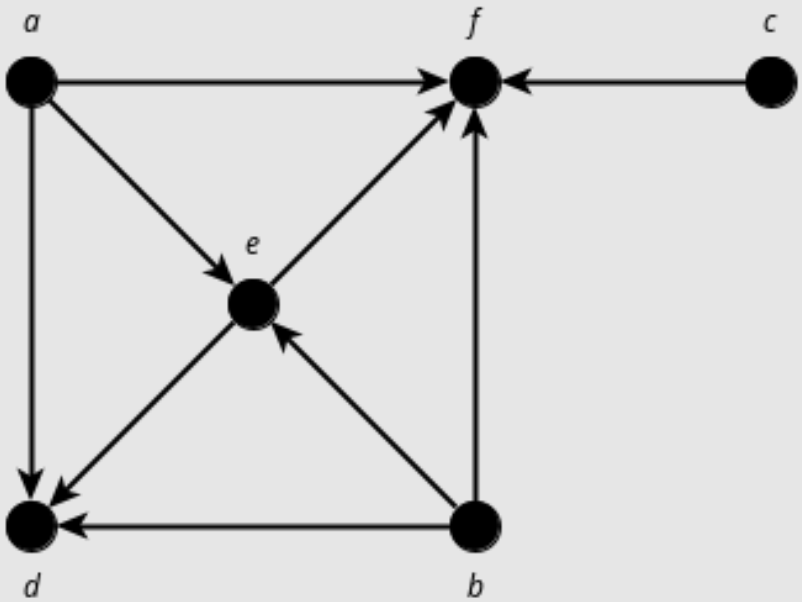
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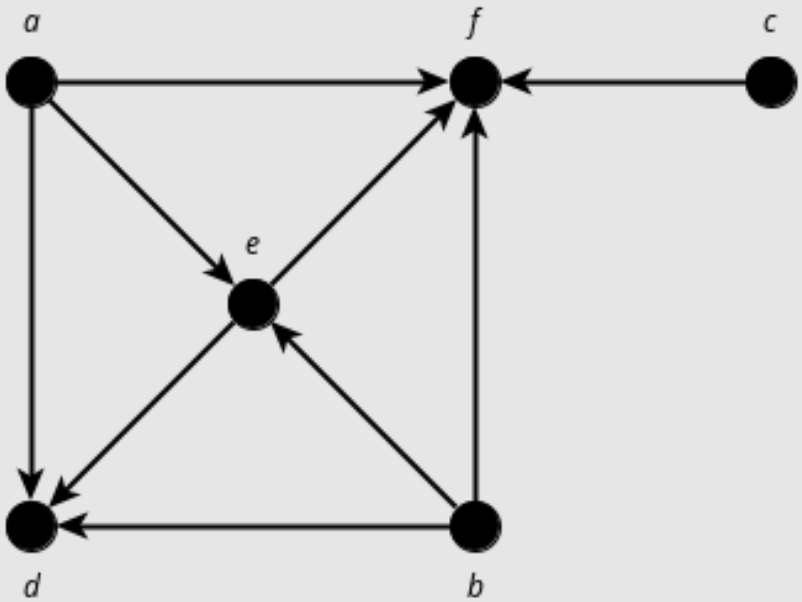
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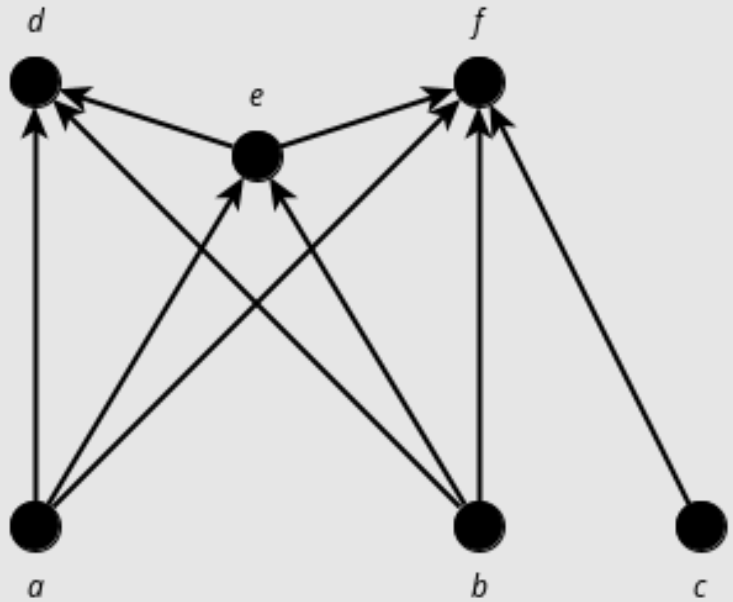
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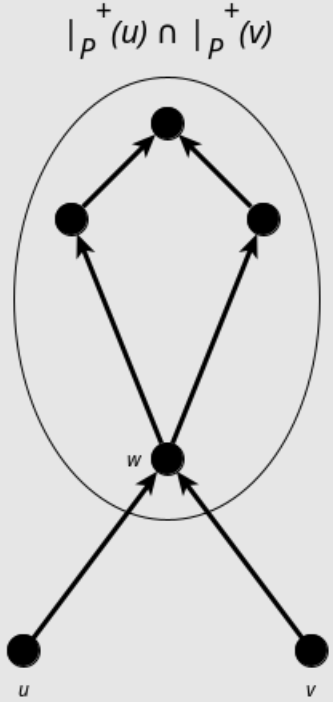
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That is, for every pair $u, v \in C$:

- $I^-_{\mathcal{P}}(u) \cap I^-_{\mathcal{P}}(v) \neq \emptyset \implies \exists w \in C, I^-_{\mathcal{P}}(w) = I^-_{\mathcal{P}}(u) \cap I^-_{\mathcal{P}}(v)$
  (w is a maximum of $I^-_{\mathcal{P}}(u) \cap I^-_{\mathcal{P}}(v)$);
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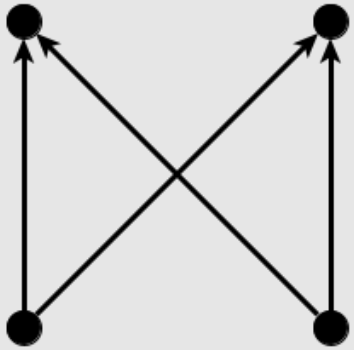
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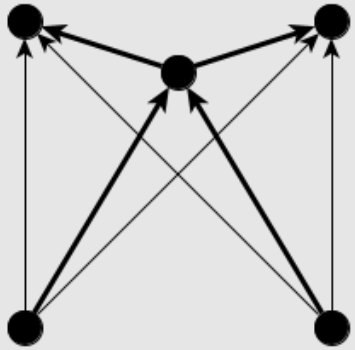
IIC-comparability: comparability graphs with posets that are *IIC*.

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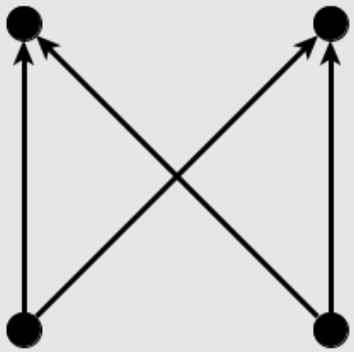
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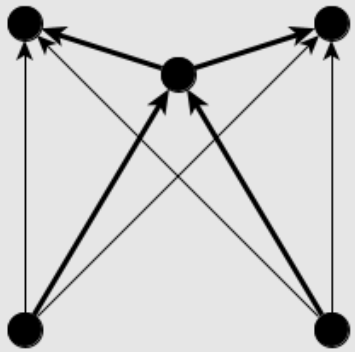




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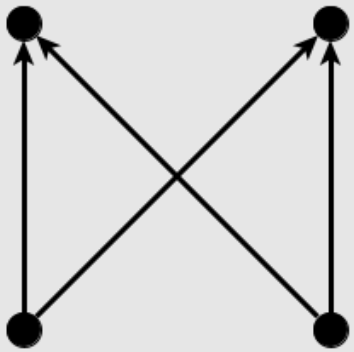


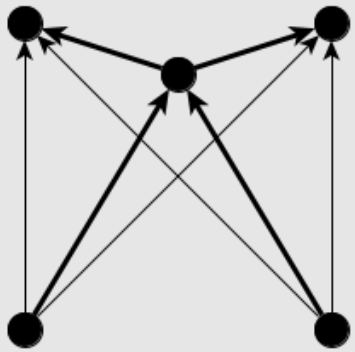


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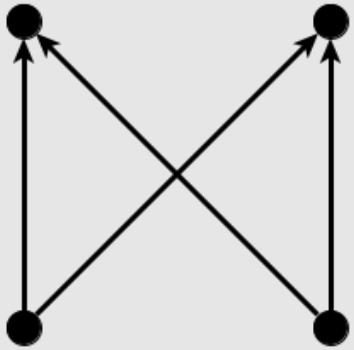


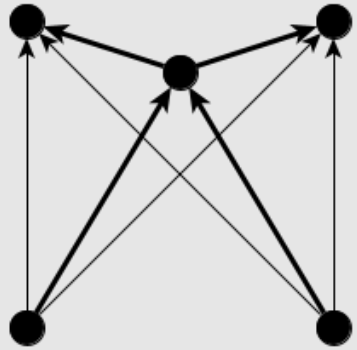


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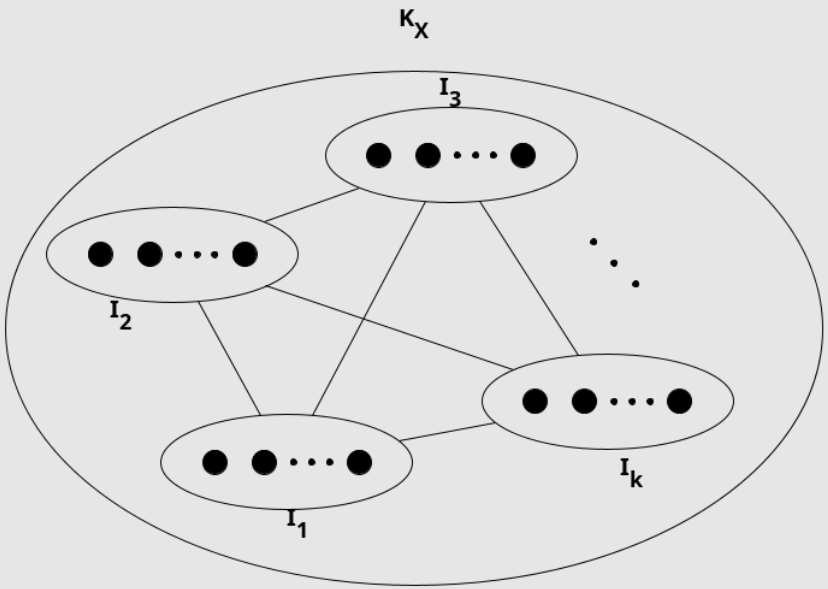
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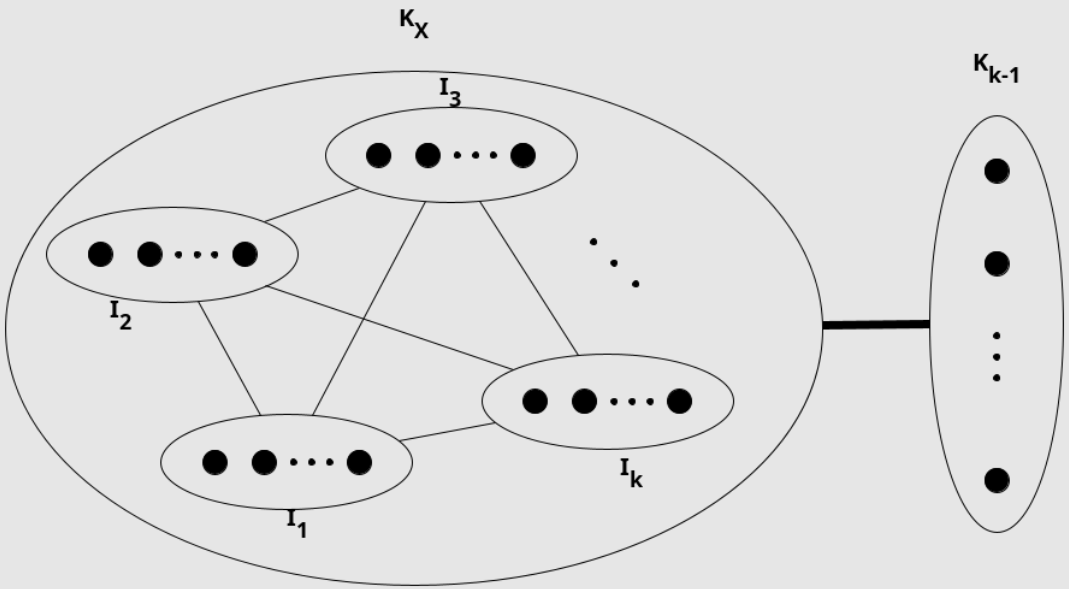
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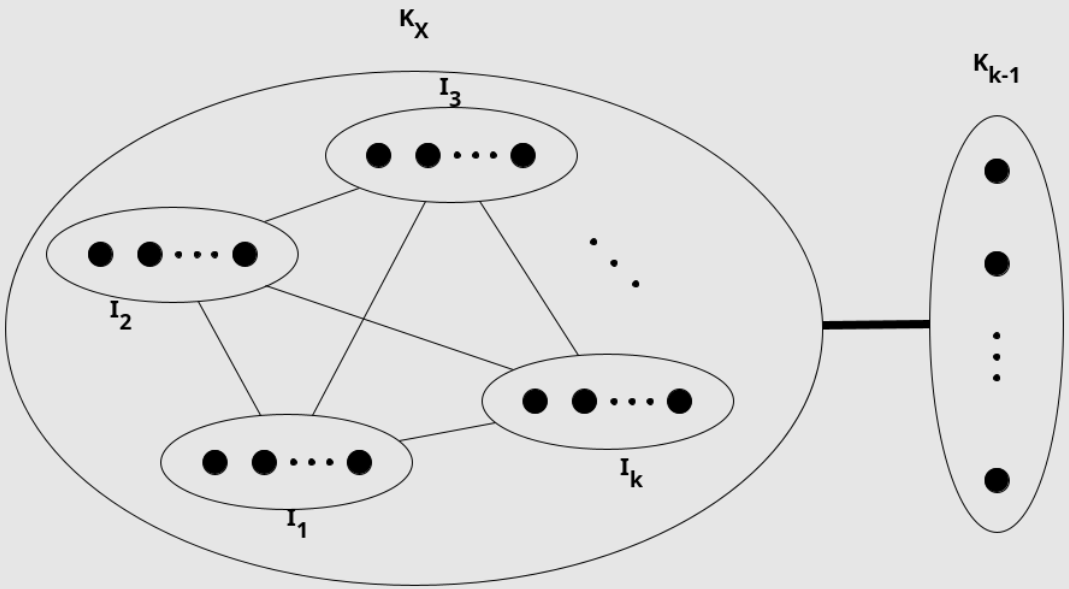
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Note that if $X = (2, 2)$, K_X is a C_4 and S_X is a 4-*wheel*.

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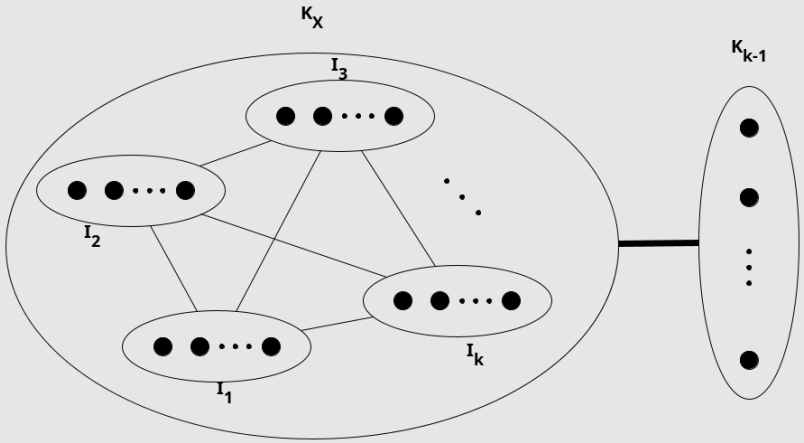
In an IIC-comparability graph, every induced subgraph K_X is a subgraph of some induced S_X such that for every vertex $w \notin V(S_X)$ adjacent to some vertex of K_{k-1}, it follows that $N(w) \cap V(K_X)$ is a transversal of the cliques of K_X, that is, it contains at least one of the parts of K_X.

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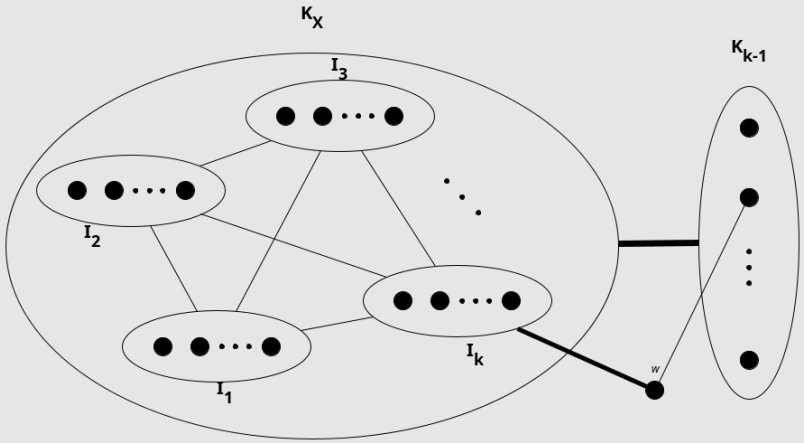
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- The complexity of recognizing biclique graphs of bipartite graphs (KB(bipartite)) is also open.
- The recognition problem of comparability graphs is polynomial, but the complexity of recognizing *IIC*-comparability graphs is still open.
- The complexity of recognizing square graphs (in general) is $\mathcal{NP}$-complete. The complexity of recognizing square graphs of *IIC*-comparability graphs is still open.

Table I

class $\mathcal{A}$	$KB(G)$, $G \in \mathcal{A}$	**class** $KB(\mathcal{A})$	**complexity**
complete	$L(G)$ [1]	L(complete) [1]	$\mathcal{P}$ [15, 17, 18]
tree	$(G - leaves(G))^2$ [1]	$(\text{tree})^2$ [1]	$\mathcal{P}$ (linear) [16]
path (P_n)	$\emptyset$, for $n = 1$ K_1, for $n = 2$ $(P_{n-2})^2$, for $n > 2$ [1]	$(\text{path})^2$ [1]	$\mathcal{P}$ (linear) [16]
half (with $2n$ vertices) [5]	K_n	complete	$\mathcal{P}$ (linear)
star	K_1	K_1	$\mathcal{P}$ (constant)
bistar	K_2	K_2	$\mathcal{P}$ (constant)
(K_3, P_4)-free graphs	$\mathcal{C}_{>1}(G)K_1$	nK_1 [4, 3]	$\mathcal{P}$ (constant)
$(K_3, \text{co-}P_3)$-free graphs	K_1	K_1	$\mathcal{P}$ (constant)
caterpilar	$(G - leaves(G))^2$	$(\text{path})^2$	$\mathcal{P}$ (linear) [16]

Table II

class $\mathcal{A}$	$KB(G)$, $G \in \mathcal{A}$	**class $KB(\mathcal{A})$**	**complexity**
cycle (C_n)	K_1, for $n = 4$ $(C_n)^2$, for $n \neq 4$ [1]	$(\text{cycle})^2 - K_4 + K_1$ [1]	$\mathcal{P}$ [6]
$\mathcal{G}_k$, for $k \geq 5$	$(G - \mathit{leaves}(G))^2$ [1]	$(\mathcal{G}_k)^2$, for $k \geq 5$ [1]	$\mathcal{P}$, for $k \geq 6$ $\mathcal{NP}$-complete, for $k = 5$ [6]
ICB	**OPEN**	$\subset (\mathit{IIC} - \mathit{PG})^2$ [4, 3, 2]	*
IBG	**OPEN**	$\subset (\mathit{IIC} - \mathit{PG})^2$ [4, 3] $\subset K_{1,4}$-free co-*CG* [1]	**OPEN**
PIB	$(L(S(G))^2$ [1]	$(L(\mathit{PIB}))^2$ [1]	**OPEN**
PIB-ASG	$(L(S(G))^2$ [1]	1-*PIG* [1]	$\mathcal{P}$ [1]
HIB	$K(G^2)$ [13]	$\subset \mathit{PIG} \cap (L(\mathit{PIB}))^2$ [1, 9, 12, 14]	**OPEN**
DFCB	$K(G^2)$ [13]	$\subset$ Dually Chordal	**OPEN**
HONHG	$K(G^2)$ [13]	—	**OPEN**

Table III

class $\mathcal{A}$	$KB(G)$, $G \in \mathcal{A}$	**class** $KB(\mathcal{A})$	**complexity**
BBHGD	**OPEN**	*CHBDI* [13]	**OPEN**
NSSG	—	—	$\mathcal{P}$ [7, 10]
threshold graphs	—	—	$\mathcal{P}$ [7, 10]
K_3-free graphs	$(KB_m(G))^2$ [8, 11]	$\subset \mathcal{G}^2$ [8, 11]	**OPEN**
bipartite	$(KB_m(G))^2$ [8, 11]	(*IIC*-comparability)^2 [8, 11] Characterization [13]	**OPEN**
$\mathcal{G}$	—	Characterization [13]	**OPEN**

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